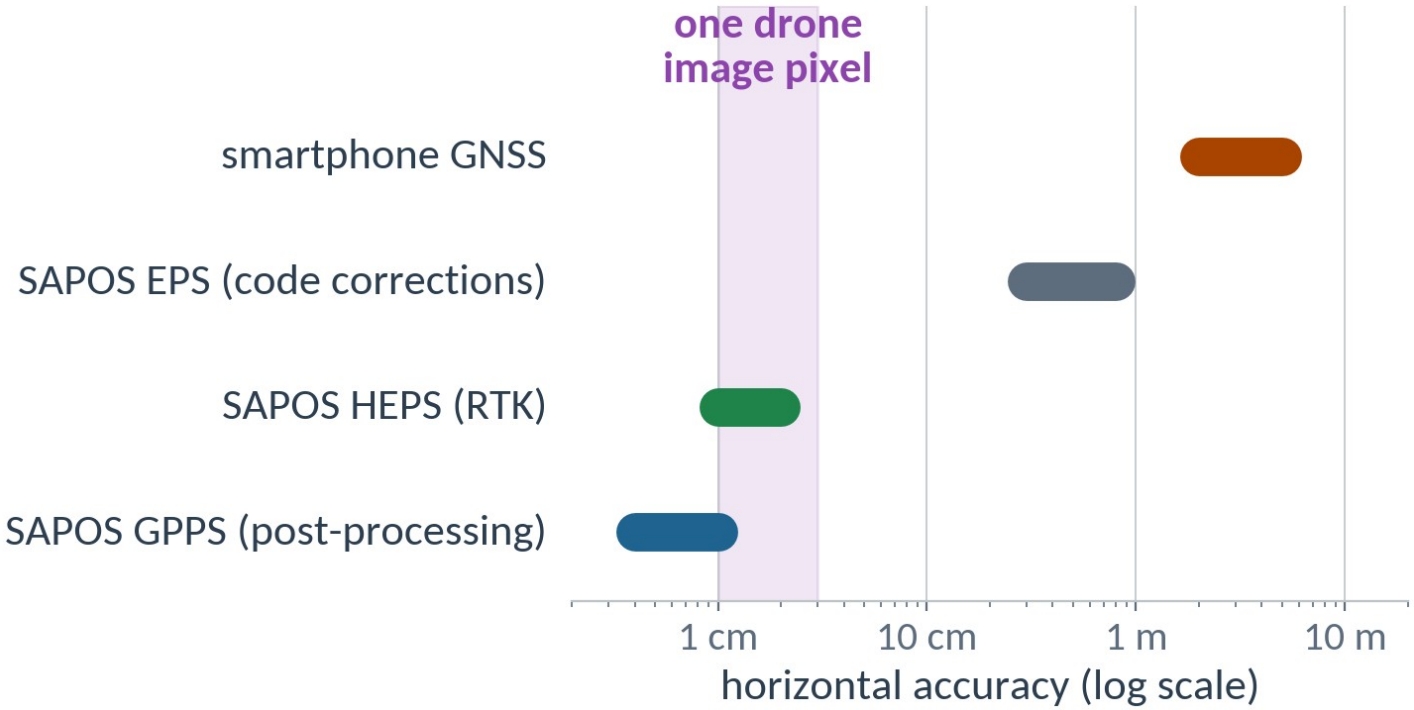
GNSS, RTK and SAPOS

Centimetres for drone mapping · geo0260



Why centimetres?



- Drone maps have 1–3 cm pixels
- Ground control points must be better than one pixel
- **→ RTK with SAPOS HEPS**

Pause: your phone shows ±4 m. What does that number mean?

Today

- **GNSS:** how a receiver finds its position, and why it is off by metres
- **DGNSS / RTK:** a base station removes the errors; carrier phase gives centimetres
- **SAPOS:** the German network of reference stations, and what you get from it
- **Heights:** from the GNSS height to a height above NHN
- **In the field:** a good RTK measurement and what the receiver exports



1 GNSS

How a receiver finds its position

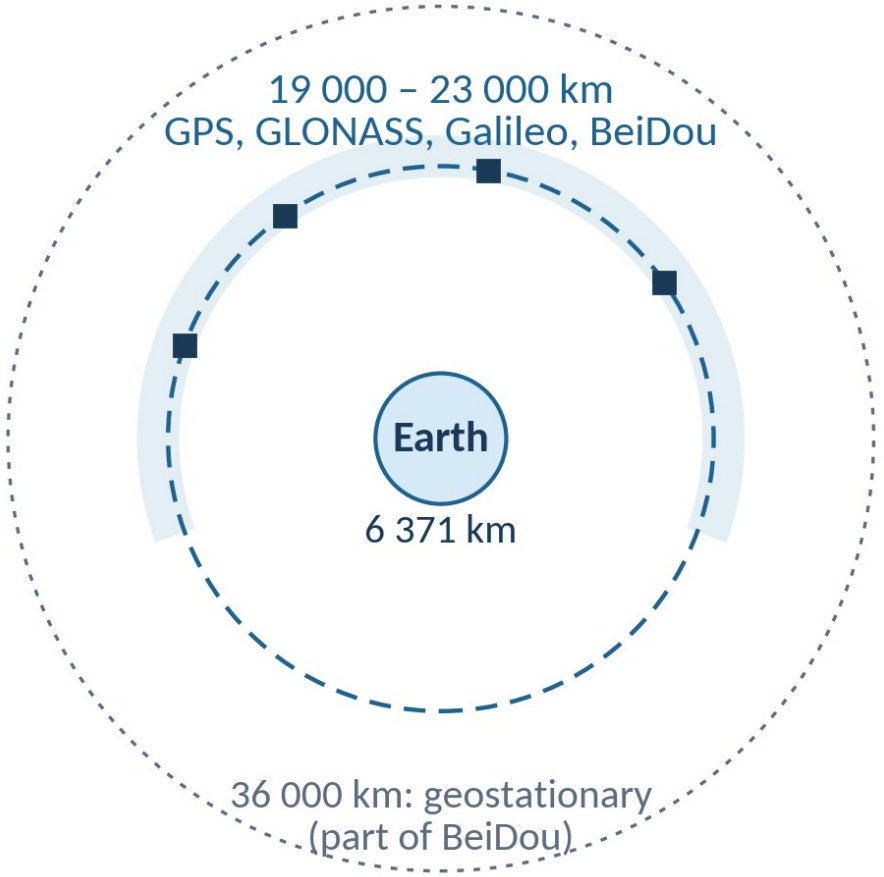
Why cm? | **GNSS** | RTK | SAPOS | Heights | Field



Four satellite systems

system	operator	height	satellites	signals
GPS	USA	20 180 km	≈ 31	L1, L2, L5
GLONASS	Russia	19 130 km	≈ 24	G1, G2, G3
Galileo	EU	23 222 km	≈ 27	E1, E5a/b, E6
BeiDou	China	21 150 km	≈ 45*	B1, B2, B3

- GNSS = all systems; GPS is one of them
- Your receiver uses all four, on two frequencies

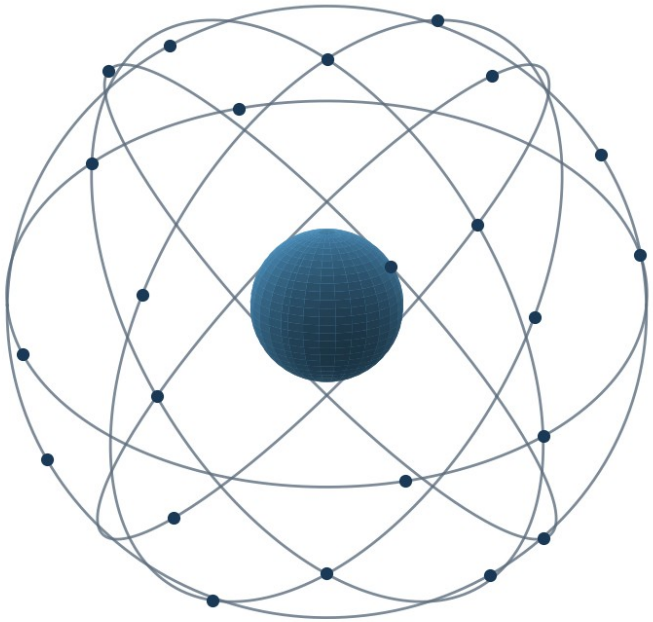


Wikipedia: Satellite navigation · GPS · GLONASS · Galileo · BeiDou

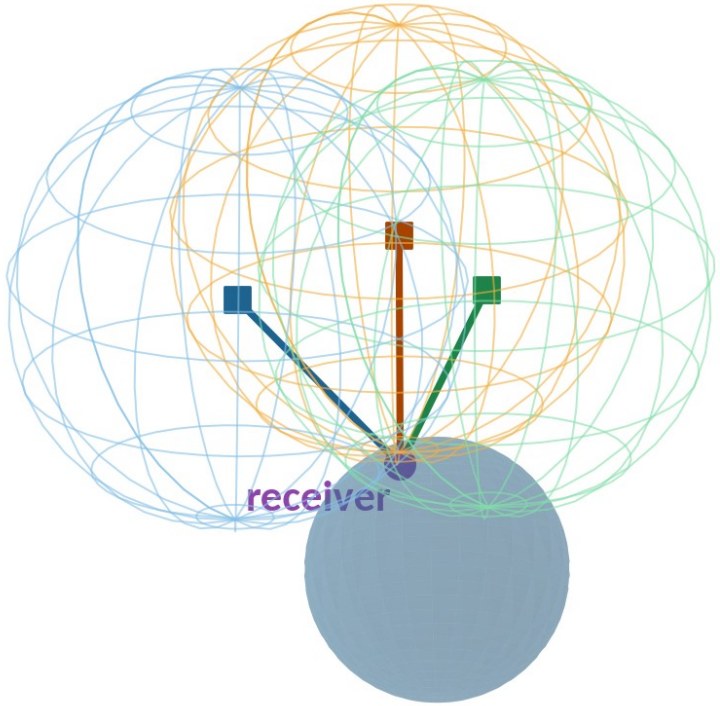
Satellite counts rounded, status 2026-10; *BeiDou incl. geostationary satellites.

Ranging in 3D: spheres around satellites

GPS: 6 orbit planes at 55°, 4 Earth radii out



one sphere per satellite: they meet in the receiver



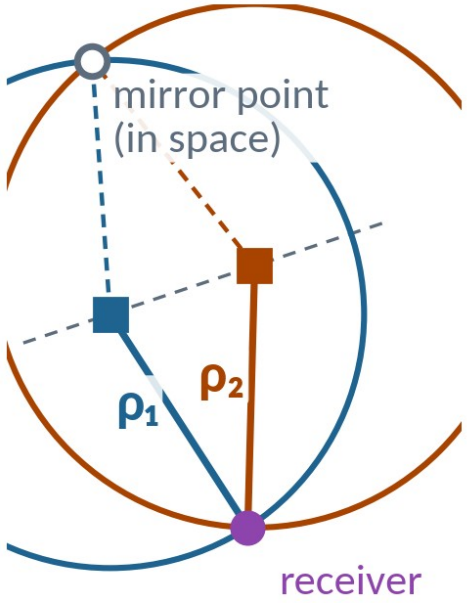
not to scale

1 range → sphere · 2 → circle · 3 → 2 points (one in space) · 4th satellite: clock

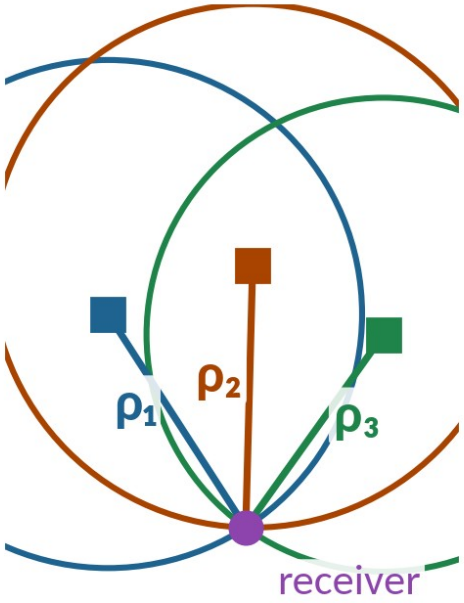
Wikipedia: GPS satellite constellation · True-range multilateration

Positioning by ranging

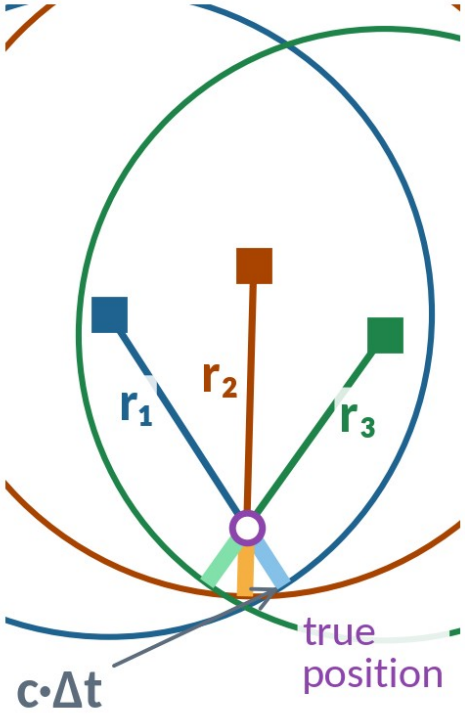
2 ranges: 2 points



3rd range picks one



clock 1 µs late: no common point



pseudorange ρ = r + c · Δt · unknowns: X, Y, Z, Δt

Pause: why four satellites in 3D?

Wikipedia: Pseudorange · Trilateration · GNSS positioning calculation

Try it: a position from pseudoranges

Self-study

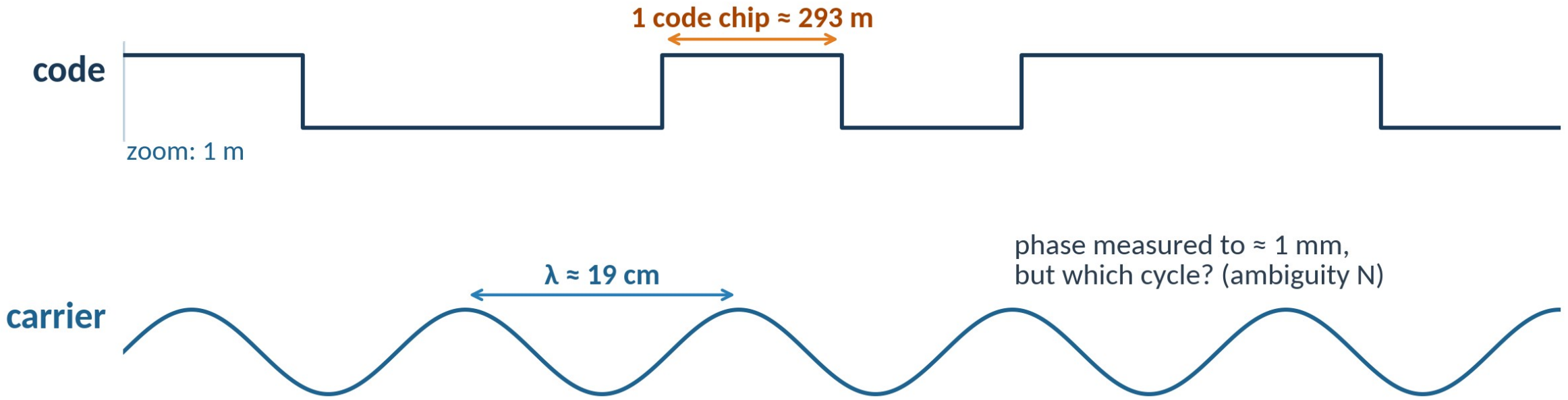
```
# sat: (n, 3) satellite positions [m], rho: (n,) pseudoranges [m]
x = np.zeros(4)           # unknowns X, Y, Z, c*dt; start: Earth's centre
for _ in range(10):       # Gauss-Newton
    r = np.linalg.norm(sat - x[:3], axis=1)
    A = np.c_[(x[:3] - sat) / r[:, None], np.ones(len(r))]
    dx = np.linalg.lstsq(A, rho - (r + x[3]), rcond=None)[0]
    x += dx
    if np.linalg.norm(dx) < 1e-3:
        break
print(x[:3], x[3] / 299_792_458)   # position, clock error [s]
```

6 satellites, ±2 m noise, clock 100 µs off → **5 iterations, error 2.3 m, clock found (99.99 µs)**

Wikipedia: GNSS positioning calculation · Gauss–Newton algorithm



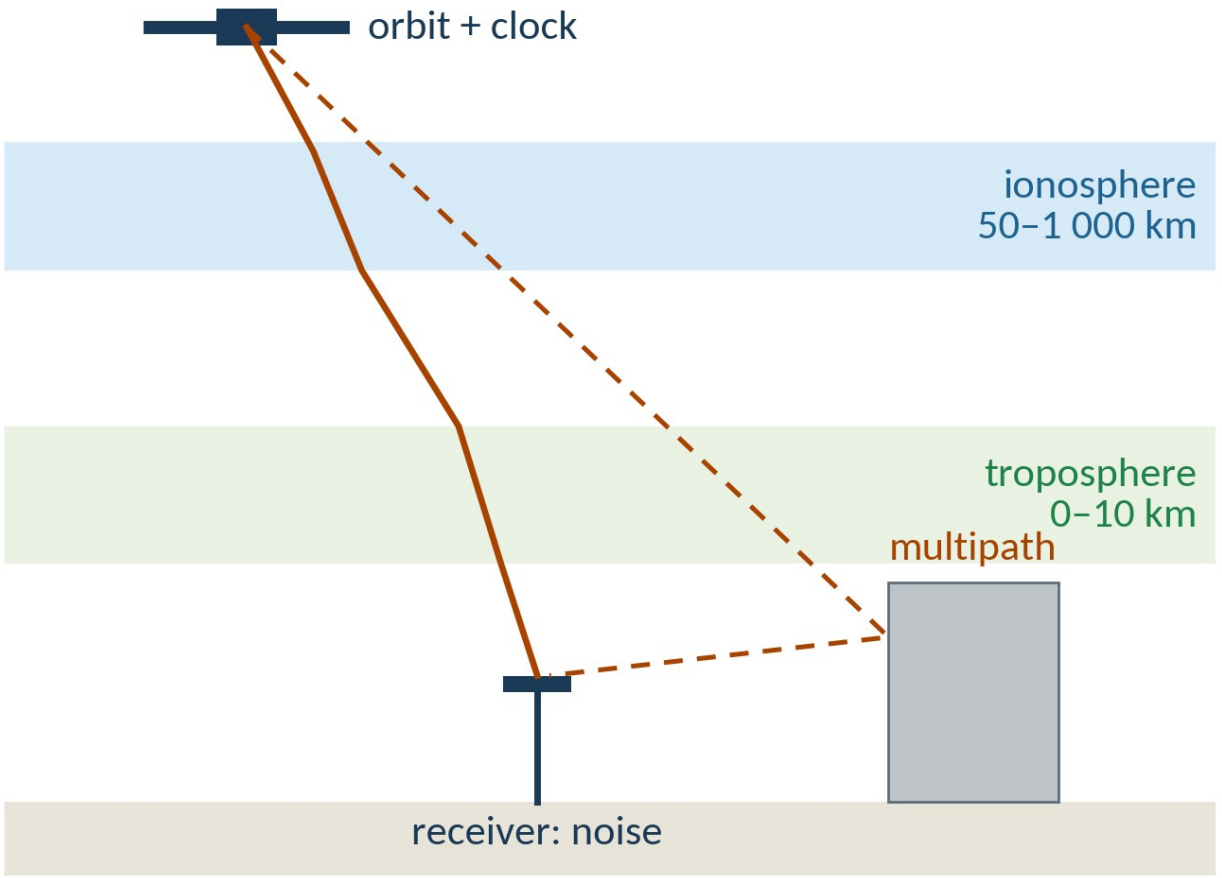
Code and carrier



	measures	precision	used by
code	time shift of the code	≈ 0.3–3 m	phones, navigation, DGNSS
carrier	phase of the wave + N cycles	≈ 1–2 mm	RTK, surveying

Wikipedia: GPS signals · Carrier phase in RTK · Gold code

Where the metres come from



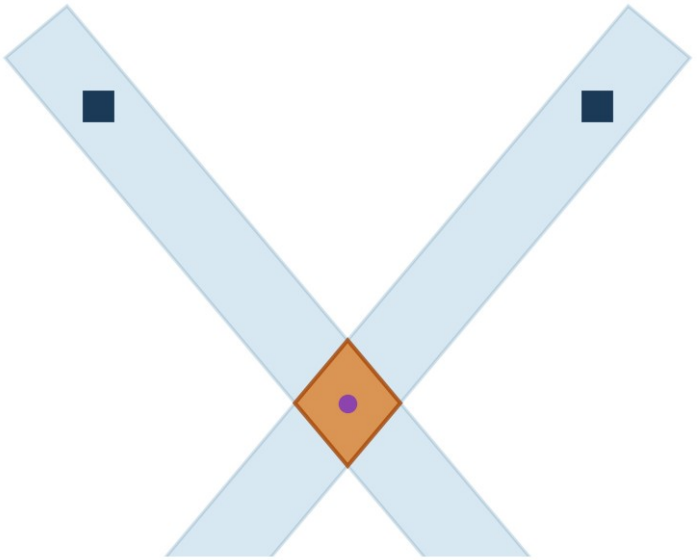
source	range error
satellite orbit + clock	≈ 0.5–2 m
ionosphere	≈ 1–5 m
troposphere	≈ 0.2–0.5 m
multipath	≈ 0.2–2 m
receiver noise	≈ 0.1–0.5 m

- **Same for receivers nearby:** orbit, clock, atmosphere
- **Local:** multipath, noise

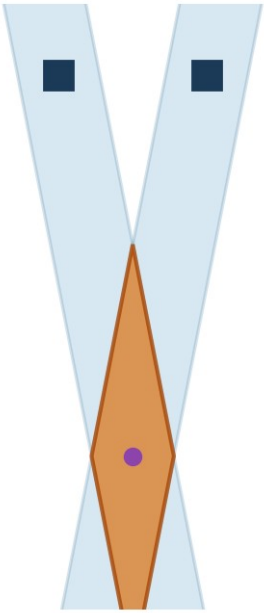
Wikipedia: Error analysis for GNSS · Ionosphere · Multipath propagation

Geometry matters: DOP

satellites spread: small DOP



satellites close together: large DOP



same range error (blue bands) · position uncertainty (orange)

position error ≈ DOP × range error PDOP < 2 good · 2–4 acceptable · > 4 poor

Wikipedia: Dilution of precision

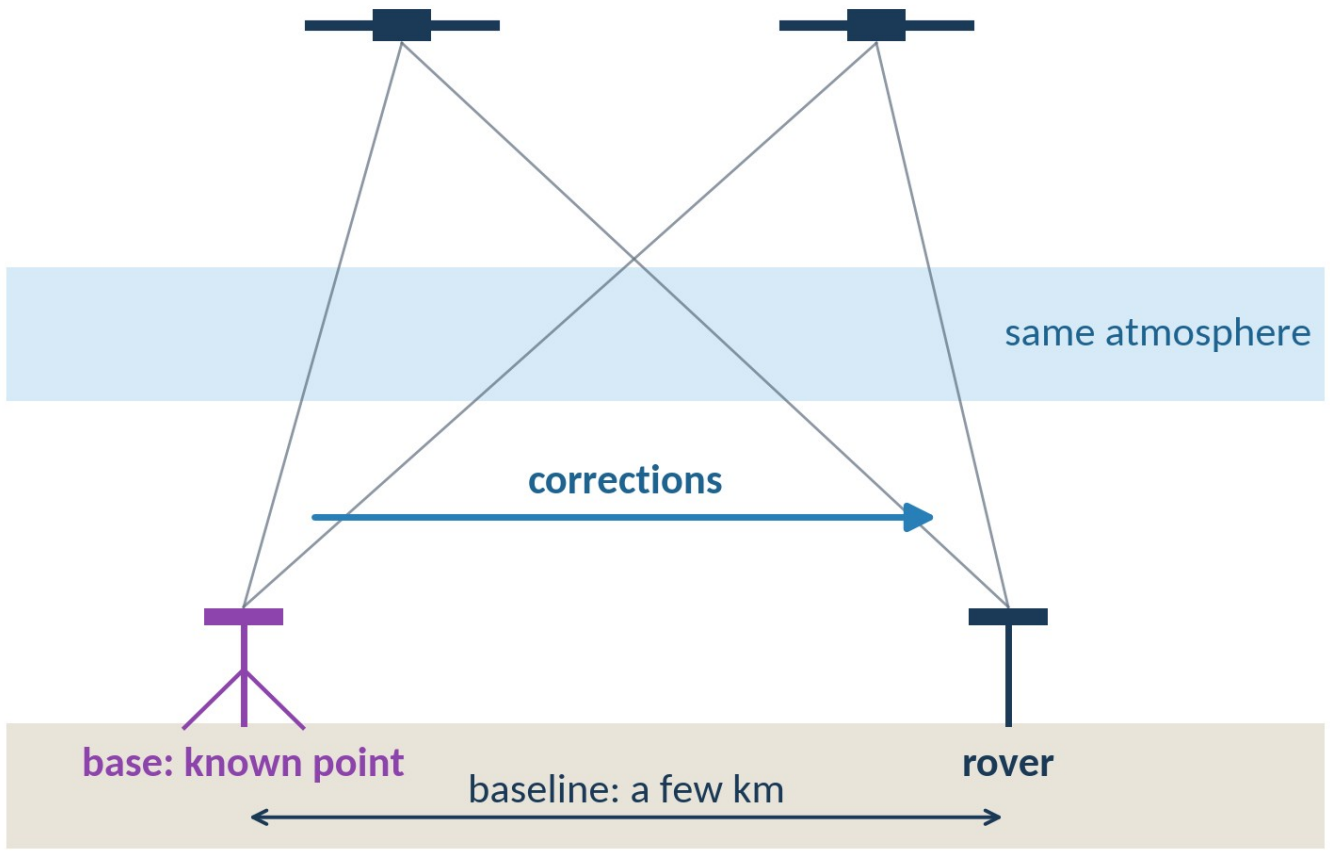
2 DGNSS and RTK

Removing the errors with a base station

Why cm? | GNSS | **RTK** | SAPOS | Heights | Field



The differential idea



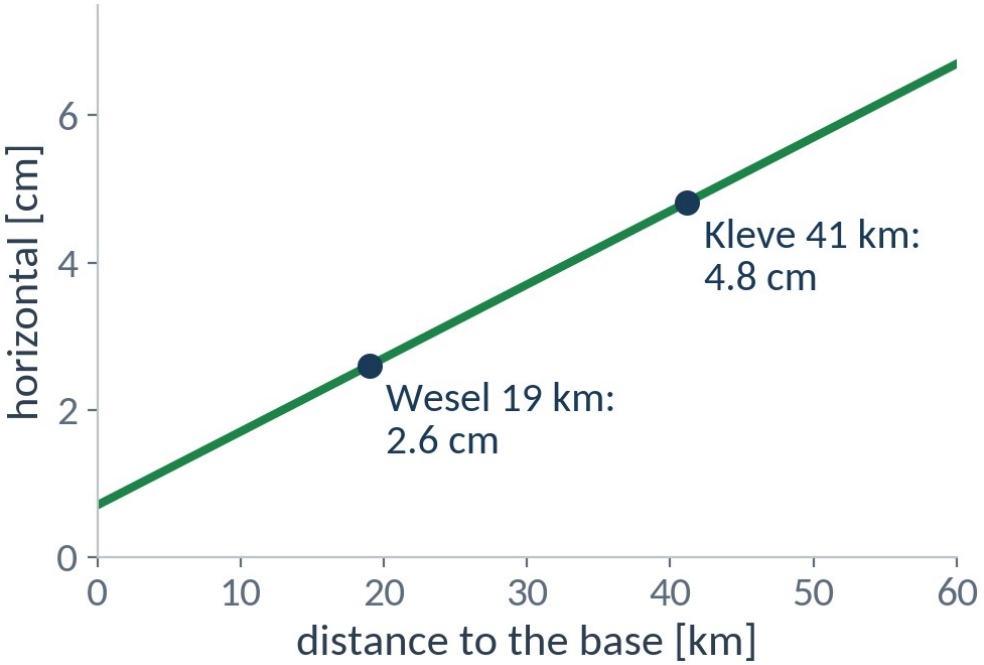
- Base on a known point measures the errors
- Sends corrections to the rover
- Errors cancel if both are close

Pause: what must the base know?

Wikipedia: Differential GNSS · Baseline (surveying)

DGNSS and RTK

	DGNSS	RTK
corrects	code ranges	carrier phase
accuracy	0.3–1 m	1–2 cm
receiver	any	dual-frequency
base	up to 100s km	≲ 20 km, or network
SAPOS	EPS	HEPS

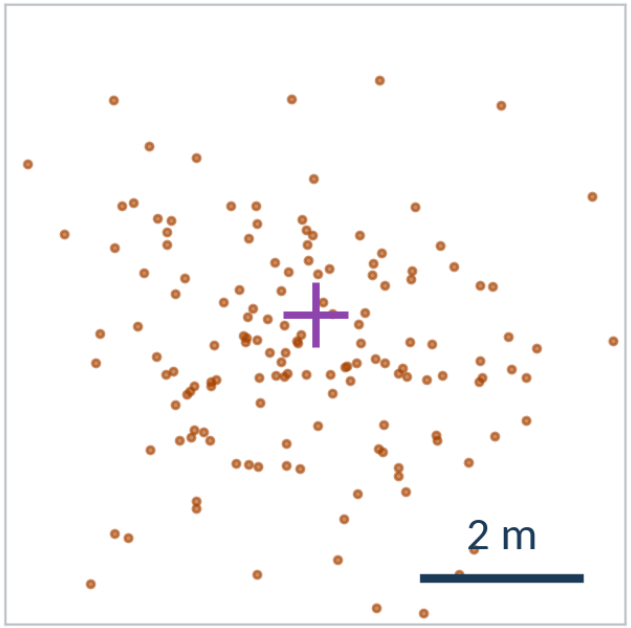


RTK spec Reach RS2+: 7 mm + 1 ppm
(height 14 mm + 1 ppm)

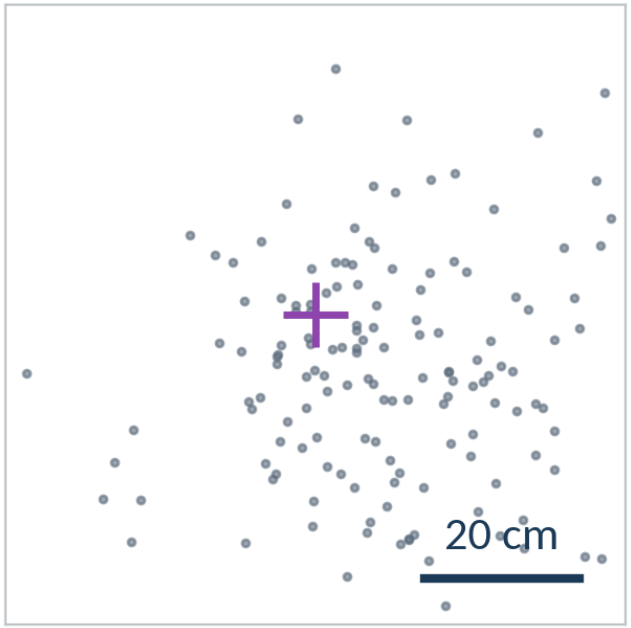
Wikipedia: Differential GNSS · Real-time kinematic positioning

Single, float, fix

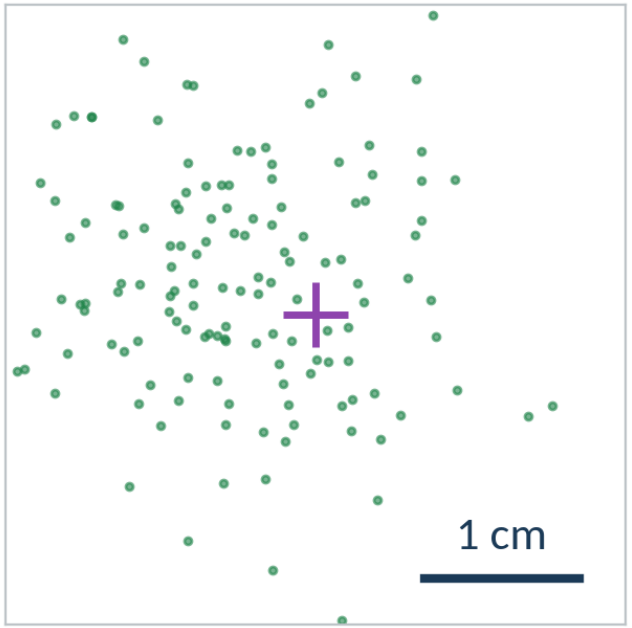
SINGLE: metres



FLOAT: decimetres



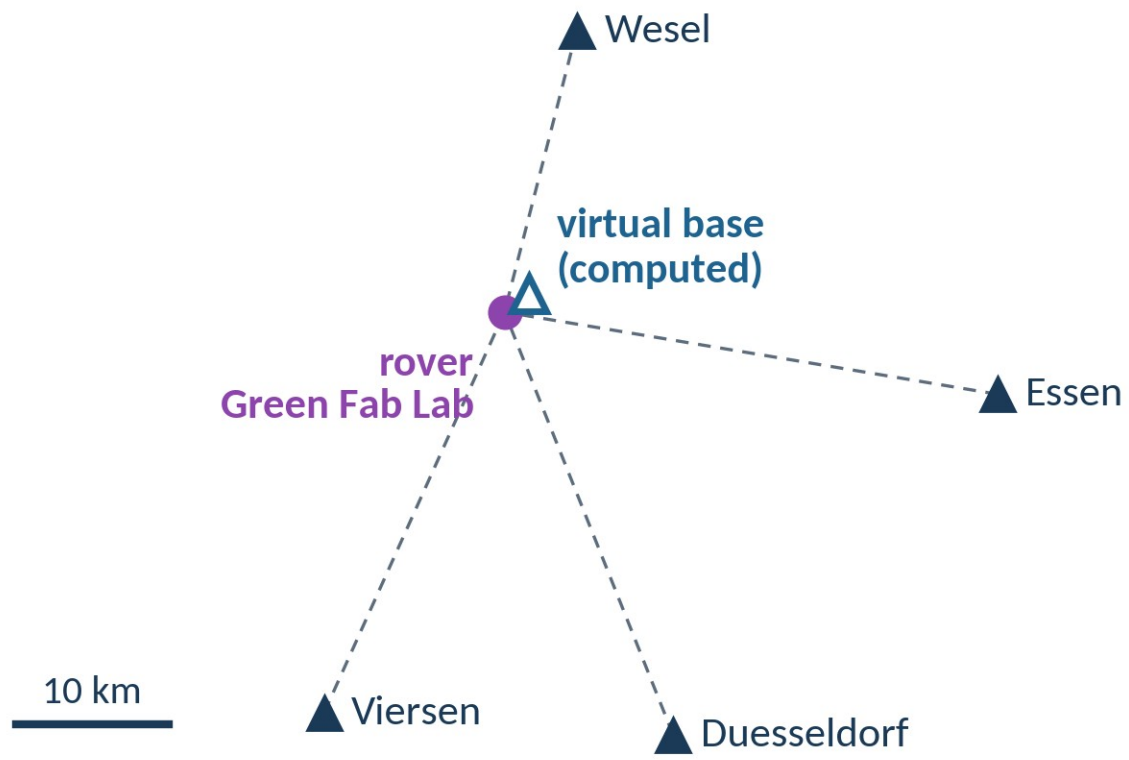
FIX: centimetres



Only FIX for ground control points. Wait for it, keep the pole still, check the RMS, measure twice.

Wikipedia: Real-time kinematic positioning · GNSS augmentation

Network RTK: a virtual base next to you



1 rover sends its rough position

2 network models the errors from the stations around

3 server sends corrections for a virtual base next to the rover

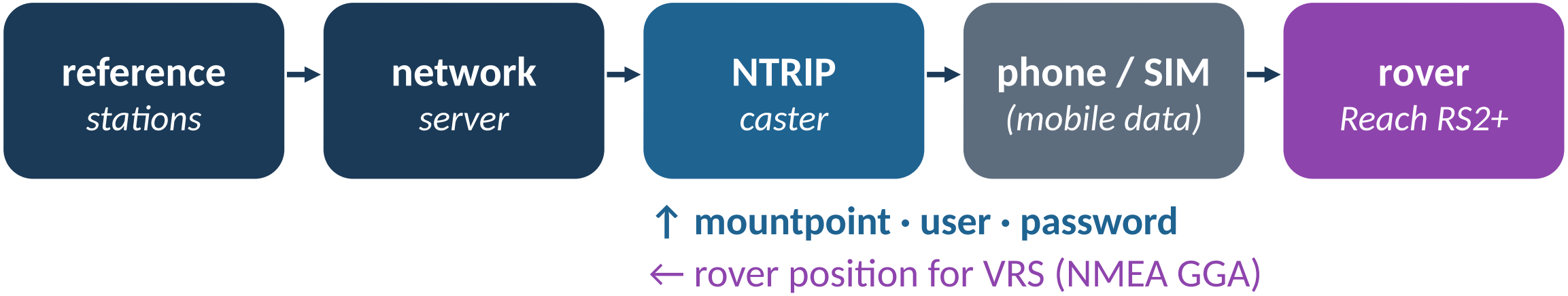
→ same accuracy everywhere in NRW

Wikipedia: Virtual reference station · Real-time kinematic positioning

SAPOS stations: Geobasis NRW, station logs (Stationsparameter.zip, GPPS archive), dl-de/zero-2-0

NTRIP: corrections over the internet

Self-study



- **Caster:** server with several correction streams (mountpoints)
- **Format:** RTCM 3, about one message per second
- **No mobile data → no corrections → no FIX**

Wikipedia: NTRIP · RTCM SC-104 · NMEA 0183

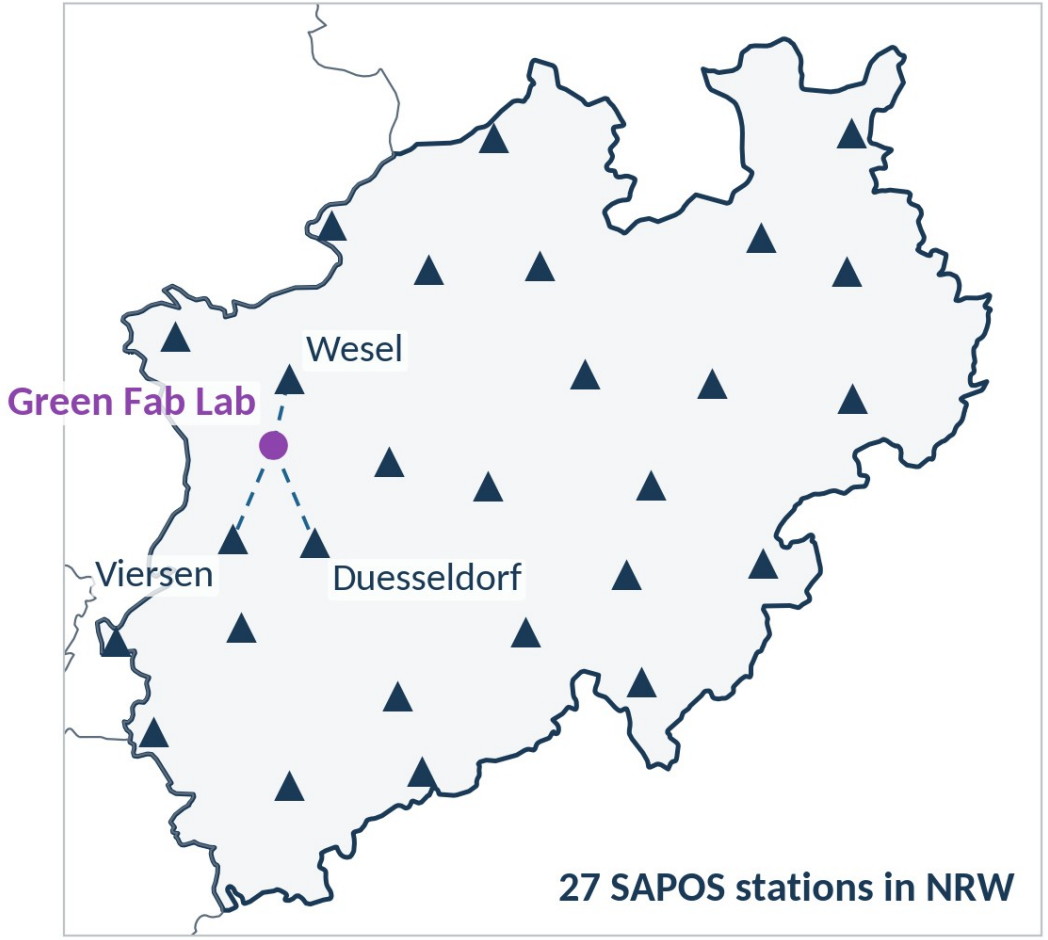
3 SAPOS

The reference network of the German survey offices

Why cm? | GNSS | RTK | **SAPOS** | Heights | Field



SAPOS: the German reference network



- Run by the state survey offices (AdV)
- > 270 stations in Germany
- NRW: 27 stations, ≈ 30 km apart
- GPS, GLONASS, Galileo, BeiDou
- **Official frame: ETRS89**

Wikipedia: SAPOS (de) · ETRS89

SAPOS stations: Geobasis NRW, station logs (Stationsparameter.zip, GPPS archive), dl-de/zero-2-0 · Borders: Natural Earth (public domain)

SAPOS services

service	mode	horizontal	height	format
EPS	real time, code	0.3–0.8 m	0.5–1.5 m	RTCM 2.3
HEPS	real time, RTK	1–2 cm	2–3 cm	RTCM 3
GPPS	post-processing	≤ 1 cm	1–2 cm	RINEX 3

- **HEPS** for ground control points (RTK)
- **GPPS** station data as RINEX: open data in NRW
- **GEPOS** one stream for Germany, in test operation

Wikipedia: SAPOS (de) · RINEX



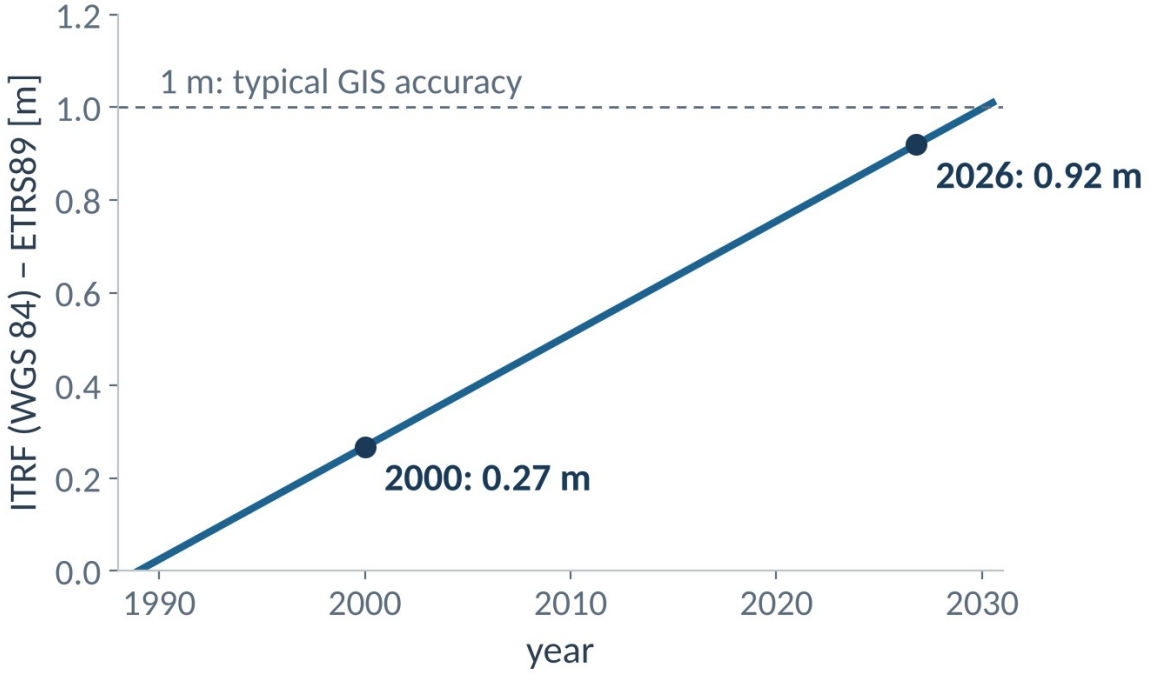
SAPOS NRW: what you enter in the app

setting	value
caster / port	sapos-nw-ntrip.de : 2101
mountpoint	VRS_3_4G_NW (4 systems, VRS)
user / password	course login, handed out on the day
send position	NMEA GGA: on (needed for VRS)
coverage	NRW only

Credentials never on slides, in notebooks or in git.



What coordinates does SAPOS give you?



- **ETRS89**, not WGS 84 (≈ 0.9 m today)
- Fixed to the Eurasian plate
- **EPSG:25832** for E, N
- **Heights:** next part

```
t = Transformer.from_crs(4258, 25832, always_xy=True)   # ETRS89 -> UTM 32N
t.transform(6.5488, 51.4981)   # E, N = 329 862, 5 708 063
```

Wikipedia: ETRS89 · Universal Transverse Mercator

Figure: geo0220 (R. Becker, CC BY-SA 4.0)

4 Heights

From GNSS to NHN

Why cm? | GNSS | RTK | SAPOS | **Heights** | Field



From a satellite fix to a height above NHN



$$H = h - \zeta$$

normal height = ellipsoidal height − height anomaly

Ellipsoidal height h is geometry, not physics

- Measured along the ellipsoid normal, above a mathematical surface
- Gravity plays no role: a calm lake is level, its h is not
- Water follows gravity, not h
- About 2 × less precise than position: satellites only above the horizon

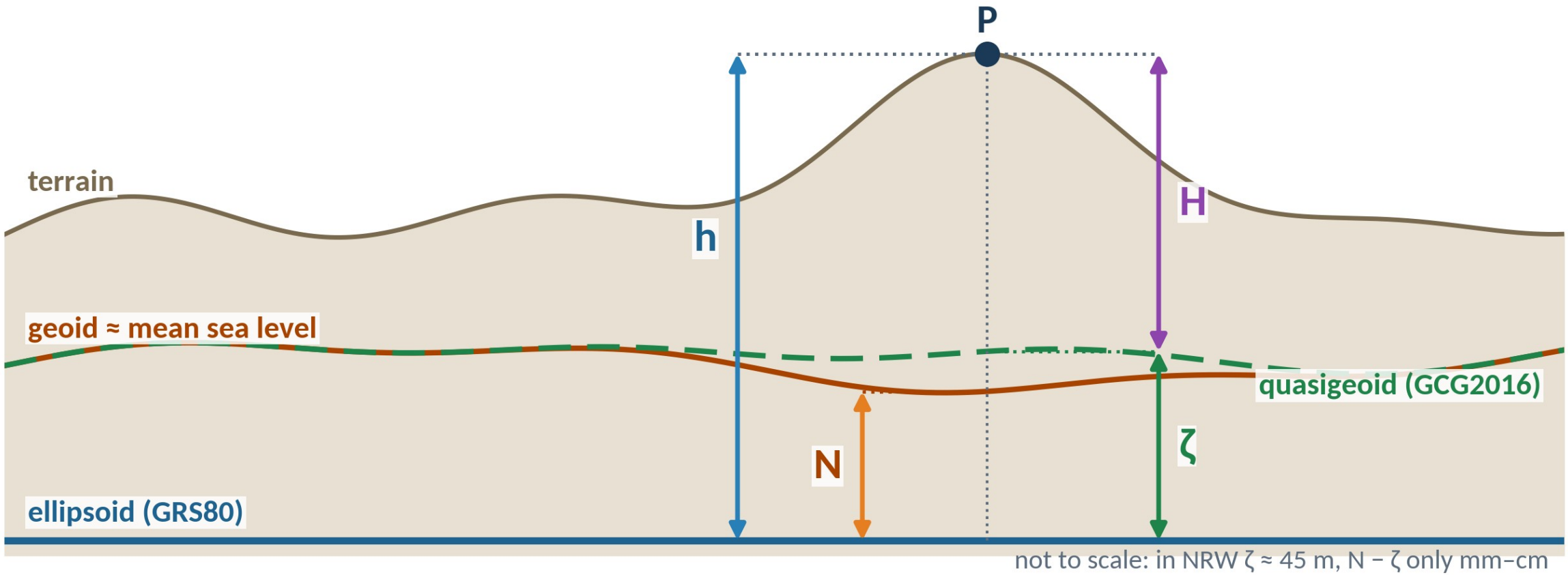
≈ 45 m

In NRW, h is about 45 m larger than the height above NHN

Wikipedia: Ellipsoidal height · Geoid



Three reference surfaces, one point



Wikipedia: Geoid · Quasigeoid (de) · Normal height · Orthometric height

Geoid or quasigeoid?

Self-study

Geoid → orthometric height

- Equipotential surface ≈ mean sea level
- $H_O = h - N$ (N: geoid undulation)
- Needs gravity inside the Earth: assumed rock density
- Physically intuitive, not exactly computable

Quasigeoid → normal height

- Auxiliary surface (Molodensky), not equipotential
- $H = h - \zeta$ (ζ: height anomaly)
- Needs gravity only at the surface: no hypothesis
- Germany since DHHN92; model GCG2016

$N - \zeta$: millimetres in the Lower Rhine plain, about a centimetre in the Sauerland

Wikipedia: Normal height · Orthometric height · Height anomaly (de)



DHHN2016: heights above NHN

Self-study

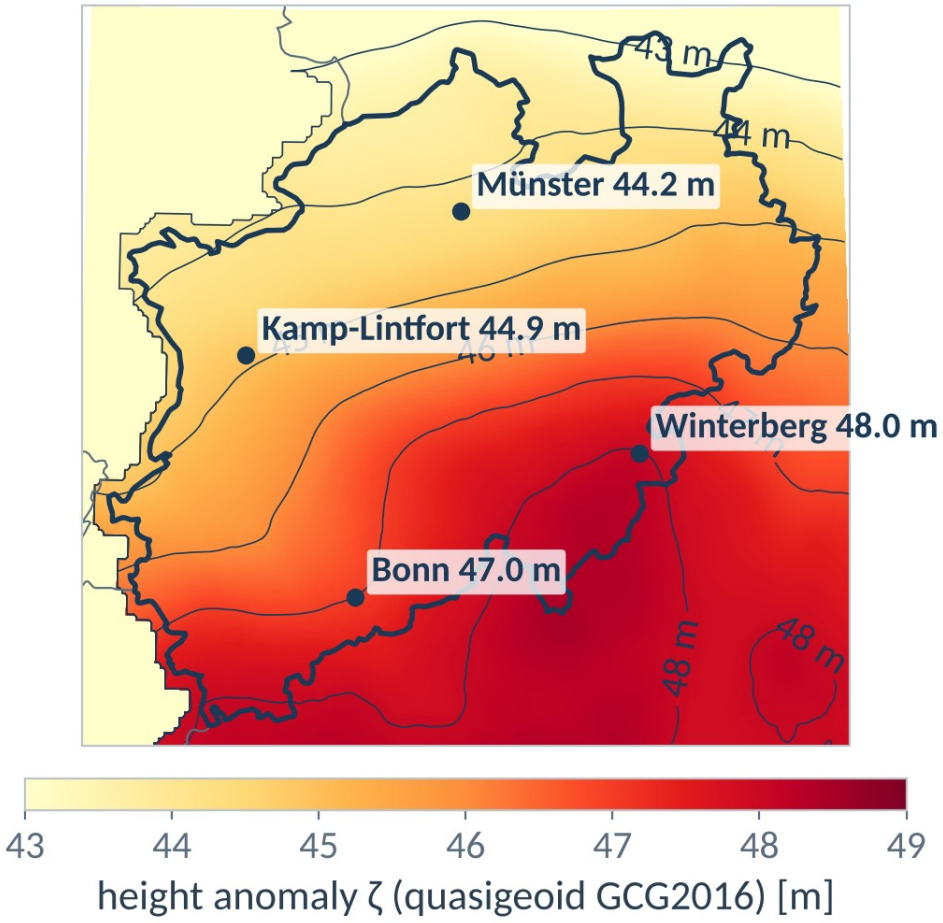
NN · DHHN12	NHN · DHHN92	NHN · DHHN2016
normal-orthometric status 100	normal heights status 160	normal heights (NRW: 2017) status 170

- NHN: zero level through the Amsterdam tide gauge (NAP)
- **Official:** “height above NHN in DHHN2016”, EPSG:7837
- DHHN92 → 2016: a few cm (mining areas: dm); convert with HOETRA2016
- **Never mix NN and NHN heights**

Wikipedia: Normalhöhennull (de) · Deutsches Haupthöhennetz (de) · Amsterdam Ordnance Datum



Worked example: the Green Fab Lab



ellipsoidal height h (GNSS)	**72.00 m**
− height anomaly ζ (GCG2016)	**44.89 m**
= height above NHN	**27.11 m**

```
# needs the GCG2016 grid
t = Transformer.from_crs(
    "EPSG:4937", "EPSG:25832+7837",
    always_xy=True)
t.transform(6.5488, 51.4981, 72)
# H = 27.11 m
```

Map: geo0220 (R. Becker, CC BY-SA 4.0); ζ: GCG2016 © BKG via cdn.proj.org

Which height is it?

Self-study

source	height relative to	minus NHN here
GNSS raw, Emlid 'Ellipsoidal height'	ellipsoid (h)	+44.9 m
SAPOS HEPS, Emlid 'Elevation'	NHN, DHHN2016	0
DGM NRW (Geobasis NRW)	NHN, DHHN2016	0
Copernicus DEM	geoid EGM2008	−0.22 m
phone, Android getAltitude()	ellipsoid (h)	+45 m
phone, 'above sea level'	global geoid (EGM)	≈ −0.1 … −0.2 m

Write the height system next to every height: h, NHN (DHHN2016) or EGM

Wikipedia: Earth Gravitational Model · Copernicus DEM



5 In the field

A good RTK measurement

Why cm? | GNSS | RTK | SAPOS | Heights | **Field**



A good RTK measurement

Before

Receiver charged, mobile data

SAPOS login tested

Project CRS: EPSG:25832 + DHHN2016

Antenna (pole) height entered

At every point

Open sky: away from walls, cars, trees

Pole vertical (bubble)

Wait for FIX, then average ≈ 30 s

Measure again 15–30 min later

Note name, time, anything special



What the CSV tells you

columns	meaning
Name, Code	point name and code
Easting, Northing, Elevation	in the project CRS (EPSG:25832 + DHHN2016)
Longitude, Latitude, Ellipsoidal height	ETRS89, h
Easting / Northing / Elevation RMS	precision estimate [m]
Solution status	FIX / FLOAT / SINGLE
Averaging start, end, Samples	how long the point was measured
PDOP, GPS/GLONASS/Galileo/BeiDou Satellites	geometry, number of satellites
Mount point, Baseline, Antenna height	correction source, distance, pole

Next: CSV → GeoPandas: keep FIX, check RMS, EPSG:25832, GeoPackage



Summary

- GNSS measures ranges; four unknowns need four satellites
- Code gives metres, carrier phase gives millimetres plus an ambiguity
- A base on a known point removes the shared errors: DGNSS, RTK
- SAPOS HEPS: network RTK in ETRS89, 1–2 cm, in all of NRW
- Heights: $H = h - \zeta$; in NRW the GNSS height is ≈ 45 m above NHN
- **Only FIX, check the RMS, measure twice, know your height**



Glossary (1/5)

Self-study

term	full name	meaning
AdV	Arbeitsgemeinschaft der Vermessungs-verwaltungen der Länder	survey offices of the states; run SAPOS
BeiDou	BeiDou Navigation Satellite System	Chinese GNSS
BKG	Bundesamt für Kartographie und Geodäsie	federal agency; publishes GCG2016
DGNSS	Differential GNSS	code corrections from a base: decimetres
DHHN2016	Deutsches Haupthöhennetz 2016	official German heights
DOP	Dilution of Precision	error ≈ DOP × range error (geometry)
DOP (map)	Digitale Orthophotos	aerial images in map geometry; not GNSS
DREF91	Deutsches Referenznetz 1991	German realisation of ETRS89 (SAPOS)

Glossary (2/5)

Self-study

term	full name	meaning
EGM2008	Earth Gravitational Model 2008	global geoid model (many drones use it)
EPS	Echtzeit-Positionierungs-Service	SAPOS real-time code service: 0.3–0.8 m
EPSG:25832	ETRS89 / UTM zone 32N	CRS of NRW: easting, northing in m
ETRS89	European Terrestrial Reference System 1989	fixed to Eurasia; ≈ 0.9 m from WGS 84
FKP	Flächenkorrekturparameter	network RTK: base + area gradients
Galileo	European GNSS	satellite system of the EU
GCG2016	German Combined QuasiGeoid 2016	quasigeoid: ellipsoidal h → height H
GCP	Ground control point	marked point, measured by RTK

Glossary (3/5)

Self-study

term	full name	meaning
GEPOS	SAPOS service in test operation	one SSR stream for all of Germany
GGA	NMEA sentence: GPS fix data	position sent to the caster (VRS)
GLONASS	Globalnaya Navigatsionnaya Sputnikovaya Sistema	Russian GNSS
GNSS	Global Navigation Satellite System	GPS, GLONASS, Galileo, BeiDou
GPPS	Geodätischer Postprocessing-Positionierungs-Service	SAPOS station data (RINEX): ≤ 1 cm
GPS	Global Positioning System	US GNSS (NAVSTAR GPS)
GRS80	Geodetic Reference System 1980	ellipsoid of ETRS89
HEPS	Hochpräziser Echtzeit-Positionierungs-Service	SAPOS network RTK: 1–2 cm

Glossary (4/5)

Self-study

term	full name	meaning
IGS	International GNSS Service	global network; site log format
MAC	Master-Auxiliary Concept	network RTK: master + differences
MSM	Multiple Signal Messages	RTCM 3 messages, all systems
NHN	Normalhöhennull	zero level of DHHN2016: “m above NHN”
NMEA	National Marine Electronics Association	text format of receiver output
NTRIP	Networked Transport of RTCM via Internet Protocol	corrections over the internet
PDOP	Position Dilution of Precision	3D position DOP: < 2 good, > 4 poor
ppm	parts per million	1 ppm = 1 mm per km of baseline



Glossary (5/5)

Self-study

term	full name	meaning
RINEX	Receiver Independent Exchange Format	text format for raw observations
RMS	root mean square	typical size of errors, residuals
RTCM	Radio Technical Commission for Maritime Services	standard formats for corrections
RTK	Real-Time Kinematic	carrier phase in real time: cm
SAPOS	Satellitenpositionierungsdienst der deutschen Landesvermessung	German reference station service
SSR	State Space Representation	corrections as error models
VRS	Virtual Reference Station	virtual base next to the rover
WGS 84	World Geodetic System 1984	frame of GPS; ≠ ETRS89 (≈ 0.9 m)



Sources and further reading (1/2)

Four satellite systems: Satellite navigation · GPS · GLONASS · Galileo · BeiDou

Ranging in 3D: spheres around satellites: GPS satellite constellation · True-range multilateration

Positioning by ranging: Pseudorange · Trilateration · GNSS positioning calculation

Try it: a position from pseudoranges: GNSS positioning calculation · Gauss–Newton algorithm

Code and carrier: GPS signals · Carrier phase in RTK · Gold code

Where the metres come from: Error analysis for GNSS · Ionosphere · Multipath propagation

Geometry matters: DOP: Dilution of precision

The differential idea: Differential GNSS · Baseline (surveying)

DGNSS and RTK: Differential GNSS · Real-time kinematic positioning

Single, float, fix: Real-time kinematic positioning · GNSS augmentation

Network RTK: a virtual base next to you: Virtual reference station · Real-time kinematic positioning

NTRIP: corrections over the internet: NTRIP · RTCM SC-104 · NMEA 0183



Sources and further reading (2/2)

SAPOS: the German reference network: SAPOS (de) · ETRS89

SAPOS services: SAPOS (de) · RINEX

What coordinates does SAPOS give you?: ETRS89 · Universal Transverse Mercator

Ellipsoidal height h is geometry, not physics: Ellipsoidal height · Geoid

Three reference surfaces, one point: Geoid · Quasigeoid (de) · Normal height · Orthometric height

Geoid or quasigeoid?: Normal height · Orthometric height · Height anomaly (de)

DHHN2016: heights above NHN: Normalhöhennull (de) · Deutsches Haupthöhennetz (de) · Amsterdam Ordnance Datum

Which height is it?: Earth Gravitational Model · Copernicus DEM

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